Question 1:

```
from scipy import stats
import numpy as np

delivery_time = np.array([
    31, 28, 35, 42, 30,
    27, 39, 33, 26, 32,
    34, 31, 38, 26, 32,
    41, 30, 36, 29, 37
])

n = len(delivery_time)

# Sample mean and sample standard deviation
sample_mean = np.mean(delivery_time)
sample_sd = np.std(delivery_time, ddof = 1) # for population ddof=0

# Degrees of Freedom
df = n - 1

# Standard Error
SE = sample_sd/ np.sqrt(n)

# 95% confidence interval
t_critical = stats.t.ppf(0.975, df)

margin_error = t_critical * SE

lower = sample_mean - margin_error
upper = sample_mean + margin_error

print("Sample Mean =",sample_mean)
print("Sample SD =",sample_sd)
print("Degrees of Freedom =",df)
print("Standard Error =",SE)
print("Margin of Error =",margin_error)
print("95% Confidence Interval =",(lower,upper))
```

```
Sample Mean = 32.85
Sample SD = 4.804329188072818
Degrees of Freedom = 19
Standard Error = 1.0742806650817192
Margin of Error = 2.248495273201745
95% Confidence Interval = (np.float64(30.601504726798257), np.float64(35.09849527320175))
```

Question 2

```
np.random.seed(42)

bootstrap_means = []

for i in range(5000):
  bootstrap_sample = np.random.choice(
      delivery_time,
      size=len(delivery_time),
      replace=True
  )
  bootstrap_means.append(np.mean(bootstrap_sample))
bootstrap_means = np.array(bootstrap_means)

# 95% percentile bootstrap CI
lower_bootstrap = np.percentile(bootstrap_means, 2.5)
upper_bootstrap = np.percentile(bootstrap_means, 97.5)

print("Bootstrap Mean =",np.mean(bootstrap_means))
print("95% Bootstrap CI =", (lower_bootstrap,upper_bootstrap))
```

```
Bootstrap Mean = 32.86674
95% Bootstrap CI = (np.float64(30.85), np.float64(35.0))
```

```
print("t-based CI =",(lower, upper))
print("Bootstrap CI =",(lower_bootstrap, upper_bootstrap))

print("t-based CI width =", lower - upper)
print("Bootstrap CI width =", upper_bootstrap - lower_bootstrap)
```

```
t-based CI = (np.float64(30.601504726798257), np.float64(35.09849527320175))
Bootstrap CI = (np.float64(30.85), np.float64(35.0))
t-based CI width = -4.496990546403492
Bootstrap CI width = 4.149999999999999
```